

Social media use and early adolescent brain structure: Findings from the Adolescent Brain Cognitive Development (ABCD) Study

Jason M. Nagata^{a,*}, Kevin Bao^a, Stuart B. Murray^b, Pierre Nedelec^a,
 Racquel A. Richardson^a, Sahana Nayak^a, Elizabeth J. Li^a, Jennifer H. Wong^a,
 Eva M. Muller-Oehring^c, Aaron Scheffler^d, Fiona C. Baker^c, Andreas M. Rauschecker^e,
 Leo P. Sugrue^e

^a Division of Adolescent and Young Adult Medicine, Department of Pediatrics, University of California, San Francisco, 550 16th Street, Box 0503 San Francisco, CA 94143, USA

^b Department of Psychiatry and Biobehavioral Sciences, University of California, Los Angeles, 10833 Le Conte Avenue, CHS - Box 951668, Los Angeles, CA 90095-1668, USA

^c Center for Health Sciences, SRI International, Menlo Park, 333 Ravenswood Ave, Menlo Park, CA 94025, USA

^d Department of Epidemiology and Biostatistics, University of California, San Francisco, 505 Parnassus Ave, San Francisco, CA 94143, USA

^e Department of Radiology & Biomedical Imaging, University of California, San Francisco, 505 Parnassus Ave, San Francisco, CA 94143, USA

ARTICLE INFO

Keywords:

Imaging
 Structural MRI
 Brain development
 Social media

ABSTRACT

Many adolescents initiate social media use during early adolescence, but the associations of early social media use with neurodevelopment have not been extensively studied. We utilized neuroimaging data from the U.S. Adolescent Brain Cognitive Development (ABCD) Study to investigate the association of social media use (hours per day) or social media addiction (Social Media Addiction Questionnaire) with brain morphology in early adolescence. We analyzed data from 7,614 participants with high-quality structural MRI and complete covariate data at Year 2 (2018–2020, ages 10–13). In addition to pre-defined cortical regions, we performed vertexwise analysis using the Fast and Efficient Mixed Effects Algorithm (FEMA), which is unbiased by arbitrary borders between atlas-based brain regions and provides higher spatial resolution. After adjusting for demographics, socioeconomic factors, genetic ancestry, non-social media screen time, and scanner features, higher average daily social media use was significantly associated with lower total cortical thickness and volume. Region-of-interest (ROI) and vertexwise analysis identified broad regions with lower cortical thickness across the prefrontal cortices, temporal lobe, occipital lobe, and parietal lobe associated with social media use and social media addiction, which overlap with key nodes of the default mode network, prefrontal executive control networks, and visual processing and attention networks. Social media addiction was not significantly associated with differences in brain morphology in ROI analysis. Our findings in a large nationwide population demonstrate that higher social media use is associated with variation in cortical morphology, but future studies are required to establish the directionality of this association.

1. Introduction

Early adolescence is a critical developmental period for physical, social, and neurocognitive maturation, including marked changes in brain structure and function (Arain et al., 2013). With the increasing

prevalence of social media use among early adolescents (McAlister et al., 2024), there is growing concern regarding its potential impact on neurodevelopment. Prior research has found associations between social media use and adverse outcomes such as cyberbullying, poor mental health, and impaired academic performance in adolescent populations

* Corresponding author at: 550 16th Street, 4th Floor, Box 0503, San Francisco, California 94143, USA.

E-mail addresses: jason.nagata@ucsf.edu (J.M. Nagata), kevin.bao@berkeley.edu (K. Bao), drstuartmurray@gmail.com (S.B. Murray), pierre.nedelec@ucsf.edu (P. Nedelec), racquel101@berkeley.edu (R.A. Richardson), snayak5@ucsc.edu (S. Nayak), elizabethli@berkeley.edu (E.J. Li), jennifer.wong@berkeley.edu (J.H. Wong), eva.muller-oehring@sri.com (E.M. Muller-Oehring), aaron.scheffler@ucsf.edu (A. Scheffler), fiona.baker@sri.com (F.C. Baker), andreas.rauschecker@ucsf.edu (A.M. Rauschecker), leo.sugrue@ucsf.edu (L.P. Sugrue).

<https://doi.org/10.1016/j.neuroimage.2026.121860>

Received 31 July 2025; Received in revised form 24 February 2026; Accepted 12 March 2026

Available online 12 March 2026

1053-8119/© 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

(Khalaf et al., 2023). Notably, high levels of social media use have been associated with variations in the developmental trajectory of cerebellar volume, although the long-term clinical implications of these findings are not yet clear (Nivins et al., 2024). Additionally, increased screen time has been linked to disruptions in neuropsychological development and heightened risk of sleep disorders and psychiatric conditions (Chen et al., 2023; Muppalla et al., 2023). In a cross-sectional study, time spent watching television was associated with lower regional cortical surface area, although effect sizes were small (Rauschecker et al., 2025). Reading, by contrast, was associated with greater regional cortical surface area; however, the relationship between social media use and these cortical metrics has not yet been studied. Given that 64% of early adolescents aged 11–12 report using social media (Nagata et al., 2025), understanding the possible interplay between social media behaviors and brain development is crucial for informing future neuroscientific and public health research (Office of the Surgeon General, 2024).

Diverse, population-based studies are essential for examining associations between adolescent behaviors and neurocognitive development while adjusting for confounding variables. The Adolescent Brain Cognitive Development (ABCD) Study is a landmark effort, following over 11,000 adolescents aged 9–10 years at baseline across 21 sites in the United States (Casey et al., 2018). With its longitudinal design and use of advanced neuroimaging techniques, including high-resolution structural magnetic resonance imaging (sMRI), the ABCD Study provides a robust framework for exploring the neurobiological and contextual foundations of adolescent brain development (Chen et al., 2023; Rauschecker et al., 2025). Importantly, the ABCD dataset includes parent-reported attention-deficit/hyperactivity disorder (ADHD) symptoms, which are commonly associated with differences in social media use and cognitive, behavioral, and brain development (Bernanke et al., 2022; Deng et al., 2025).

While existing literature has utilized functional magnetic resonance imaging (fMRI) to examine the relationship between overall screen use and adolescent brain development, few studies have focused on social media specifically (Miller et al., 2023; Nivins et al., 2024). One previous study explored these associations but focused predominantly on older adolescents (Achterberg et al., 2022). To our knowledge, three prior papers have used the ABCD dataset to explore associations between screen media use and brain structural changes. Two ABCD papers explored associations between combined screen media use (e.g., television shows/movies, videos, video games, texting, social networking sites, and videochatting): one identified links with brain structural covariation, and the other used longitudinal imaging to relate screen media use to brain structural co-development and internalizing and externalizing psychopathology (Zhao et al., 2023, 2022). In contrast, the current study focuses specifically on social media use and addiction using high-resolution cortical morphology analyses. Another ABCD paper observed variations in frontal cortex regions associated with screen time (Paulus et al., 2019). This study used group factor analysis to identify patterns of brain morphology associated with different profiles of screen media activity, including watching television or videos, playing video games, and using social media. One factor identified that was loaded on social media was associated with reduced hippocampal volume and lower inferior-temporal cortical gray matter volume. However, this factor was also associated with other forms of screen media activity, thus this analysis did not identify statistically independent associations of brain morphology with social media. Social media involves different interactive elements that distinguish it from other forms of screen media—such as watching television or videos and playing video games—and may engage different cognitive processes. Thus, social media use may be associated with different patterns of brain morphology that are distinct from these other forms, and our study aims to identify the specific relationship with social media use. Here, we expand on this initial work to consider social media addiction in addition to time and use a novel sMRI analysis of cortical morphology that provides higher spatial resolution.

2. Methods

2.1. Study population

We performed a secondary analysis of cross-sectional data collected from Year 2 (2018–2020, 10–13 years old) of the ABCD Study (6.0 release) (Jernigan et al., 2025), a longitudinal cohort study following participants from 9–10 years old at study onset through adolescence (Volkow et al., 2018). Participants were recruited in school systems across 21 sites in the United States, and school selection was informed by sex, race and ethnicity, socioeconomic status, and urbanicity to yield a population that was demographically diverse (Garavan et al., 2018). Throughout the study, participants undergo biospecimen collection, serial magnetic resonance imaging (MRI), neurocognitive assessments, and a battery of behavioral questionnaires. The University of California, San Diego institutional review board (IRB) was the IRB of record and approved the study. Informed consent was obtained from caregivers, and assent was secured from each participating adolescent.

2.2. Exclusion criteria

Exclusionary criteria were not being fluent in English, having a parent not fluent in English or Spanish, major medical or neurological conditions, gestational age under 28 weeks, birthweight under 1200g, contraindications to MRI scanning, a history of traumatic brain injury, a current diagnosis of schizophrenia, autism spectrum disorder (moderate/severe), intellectual disability, or alcohol/substance use disorder (Micheline et al., 2019). We utilized data from Year 2, where sMRI and social media addiction and social media use data were available. Of the 10,973 participants recruited, 8,059 participants had available sMRI data. The ABCD data contain a recommended imaging inclusion variable to filter for imaging quality, such as excessive motion, which is based on a combination of manual and automatic review of raw sMRI scans and FreeSurfer cortical surface reconstructions. Specific details of inclusion determination are detailed in the ABCD documentation (Hagler et al., 2019). Of the study cohort, 7,866 participants passed the recommended imaging inclusion criteria. After filtering participants for missing covariates (age, sex, race, caregiver partnership, household income, highest caregiver education, non-social media screen time, and genetic ancestry), our final sample size was 7,614 participants.

2.3. Independent variables: social media addiction and use

We assessed social media use data collected by the ABCD Youth Screen Time Survey at Year 2 (2018–2020). Participants reported the minutes and hours per day spent using social media (e.g., Instagram, Facebook, Twitter). Adolescents reported time spent on a typical weekday and weekend separately as a free-response answer, and a weighted average was calculated to determine average daily social media use $([\text{weekday average} * 5] + [\text{weekend average} * 2])/7$, utilizing a similar method from a previous study (Rauschecker et al., 2025). Average daily social media use for 14 participants was winsorized at 16 hours/day (above 99th percentile) to limit the influence of extreme outliers while retaining all observations in the analysis.

Social media addiction was assessed by the six-item Social Media Addiction Questionnaire adapted from the Bergen Facebook Addiction Scale (Andreassen et al., 2012). Validation metrics of this social media addiction scale have been assessed in a previous study (Bagot et al., 2022). Each item asks about addictive behaviors, such as “I’ve tried to use my social media apps less but I can’t” and responses were recorded on a six-item Likert scale that ranged from 1 (Never) to 6 (Very often). The questionnaire was scored by averaging across all items, omitting missing and “Decline to Answer” responses. A higher score represents greater severity of addictive behaviors. This questionnaire was only administered if participants endorsed having at least one social media account, so participants reporting no social media use were assigned a

value of 1 for their social media addiction score.

2.4. Covariates

Covariates used for adjustment were age, sex (female or male), race defined by the National Institutes of Health (Asian, Black, Native American/Alaska Native, Native Hawaiian/Pacific Islander, White, more than one race, or other), genetic ancestry, caregiver partnership, household income (<\$25,000, \$25,000-\$49,999, \$50,000-\$74,999, \$75,000-\$99,999, \$100,000-\$199,999, ≥\$200,000, Don't know, Decline to answer), and highest caregiver education (less than high school, high school diploma/GED, some college, bachelor's degree, graduate school or professional degree), MRI scanner serial number, and MRI scanner software version. Non-social media screen time, winsorized at 16 hours/day, was included as a covariate. Non-social media screen time was calculated by first winsorizing total screen time at 16 hours/day (91st percentile), and subtracting winsorized social media time from winsorized total screen time. Genetic ancestry was included as a covariate to account for population-level variation across ancestry groups that influence brain structure, such as surface area and other morphological metrics (Fan et al., 2015), ensuring that associations between social media use and neurodevelopmental outcomes are not confounded by genetic ancestry. Similar covariate adjustment strategies have been used before in previous studies (Palmer et al., 2022; Rauschecker et al., 2025). Genetic ancestry was represented as the top 10 genetic principal components using the GENESIS package obtained from the ABCD data (Gogarten et al., 2019). Family ID was added as a random effect to account for familial relationships. In sensitivity analyses, parent-reported ADHD symptoms from the Child-Behavior Checklist were included as an additional covariate.

2.5. Dependent variables: sMRI image processing

Neuroimaging data for the ABCD data is processed centrally through a standardized pipeline and described extensively in detail in other publications (Casey et al., 2018; Hagler et al., 2019). Briefly, imaging data across all sites were obtained from three tesla scanners with 32-channel head or 64-channel head/neck coils under a harmonized MRI acquisition protocol, which employs real-time motion detection for prospective motion correction during structural anatomical imaging. We utilized neuroimaging data from the ABCD data release which underwent the standard central processing pipeline, which includes FreeSurfer quality control and manual review of cortical surface reconstruction for motion, intensity inhomogeneity, white matter underestimation, pial overestimation, and magnetic susceptibility artifact. Coordination of data acquisition and centralized image processing was performed by the ABCD Data Analysis and Informatics Resource Center. Brain segmentation and cortical surface reconstruction were performed using FreeSurfer version 7.1.1 (<https://surfer.nmr.mgh.harvard.edu>) using preprocessed T1-weighted volumes as input. Measures of cortical surface area (mm²), thickness (mm), and volume (mm³) for region-of-interests (ROIs) defined by the Desikan-Killiani atlas (Desikan et al., 2006) and volumes for subcortical structures (mm³) segmented based on FreeSurfer's Aseg atlas (Fischl et al., 2002) were obtained from ABCD data. Analysis was performed on the tabulated measures from the 6.0 data release.

2.6. Statistical analysis

2.6.1. Region-of-Interest (ROI) analysis

Statistical analysis was conducted using R 4.4.2. Using the gamm4 package (<https://CRAN.R-project.org/package=gamm4>), measures of cortical morphology were set as dependent variables with either average daily social media use or social media addiction as the independent variable in a generalized additive mixed model with covariates described above. The Benjamini-Hochberg false discovery rate (FDR)

procedure was used to adjust for multiple comparisons (Benjamini and Hochberg, 1995).

2.6.2. Vertexwise analysis

We also conducted vertexwise analyses, which provide higher spatial resolution and are unbiased by arbitrary borders of atlas-defined regions and loss of information from averaging (Fürtjes et al., 2023). Previously, the use of multivariate linear mixed effects modeling on whole-brain data was limited due to heavy computational requirements. Recently, a computationally-efficient Fast and Efficient Mixed Effects Algorithm (FEMA) has made whole-brain analysis more feasible (Parekh et al., 2024). FEMA models linear effects of a predictor on each vertex while allowing for multivariate adjustment of fixed and mixed effects. Technical details of the FEMA algorithm are described by Parekh et al. (Parekh et al., 2024). Concatenated vertexwise data in .mat files were made available in the ABCD 6.0 data release and used for analysis. FEMA was implemented in MATLAB R2025a using the MATLAB package provided by Parekh et al. (https://github.com/cmig-research-group/cmig_tools/). FEMA was performed on 20,484 vertices, each with cortical surface area and thickness measurements, across both hemispheres. FEMA models included the same fixed and mixed effects employed in the ROI based analysis.

3. Results

We analyzed data from 7,614 participants (average age: 12 years (standard deviation [SD]=0.7); 53.3% male) in the ABCD Study after filtering for imaging quality and data completeness (Table 1). The mean daily hours of social media use were 0.6 (SD=1.5). The mean social media addiction score was 1.4 (SD=0.8). Social media use was significantly associated with social media addiction (Spearman's rho = 0.67, $p < 2.2e-16$), and both were significantly correlated with highest

Table 1

Sociodemographic characteristics of Adolescent Brain Cognitive Development (ABCD) Study participants (N=7,614).

Sociodemographic characteristics	Mean (SD) / (%)
Age (years)	12.0 (0.7)
Sex	
Male	53.3%
Female	46.7%
Race	
Asian	1.8%
Black	14.1%
Native American/Alaska Native	0.5%
Native Hawaiian or other Pacific Islander	0.2%
White	69.0%
More than one race	11.0%
Other (Unknown or not reported)	3.4%
Household income	
\$24,999 or less	10.0%
\$25,000 to \$49,999	12.0%
\$50,000 to \$74,999	12.9%
\$75,000 to \$99,999	13.3%
\$100,000 to \$199,999	31.7%
\$200,000 or greater	12.8%
Decline to answer	3.4%
Don't know	4.0%
Highest caregiver education	
Up to high school (No diploma)	4.2%
High school diploma/GED	8.7%
Some college	32.1%
Bachelor's degree	21.8%
Graduate school or professional degree	33.4%
Caregiver partnership	
Partnered	75.1%
Single	24.9%
Non-social media screen time (hours)	5.7 (4.1)
Social media	
Mean daily social media use (hours)	0.6 (1.5)
Mean social media addiction score	1.4 (0.8)

caregiver education and household income (Supplemental Figure A.1). Since social media use and addiction were highly correlated, their effects were estimated using separate models to avoid collinearity.

3.1. Region-of-Interest analysis of brain morphology

We utilized generalized additive mixed models with measures of brain morphology (cortical surface area, cortical thickness, cortical volume, subcortical volume) with each ROI set as the dependent variable.

Cortical surface area, cortical and subcortical volume. There was no significant association between social media use and cortical surface area or subcortical volume after FDR correction. Higher daily social media use was associated with overall lower cortical volume in the right hemisphere (100x standardized B [beta coefficient] = -1.916, SE [standard error] = 0.931, $p = 0.040$) (Table 2). After FDR correction, social media addiction was not significantly associated with cortical surface area, cortical volume, or subcortical volume in any ROI.

Cortical thickness. Higher daily social media use was associated with lower cortical thickness within several ROIs (Table 2 and Fig. 1B), predominantly located in the occipital, parietal, inferior temporal, and frontal regions. Regions with the greatest effects were the left bank of the superior temporal sulcus, left inferior parietal gyrus, right caudal middle frontal gyrus, left transverse temporal gyrus, and right superior

parietal gyrus. Right frontal cortical regions had stronger negative associations with social media use compared to left frontal regions. After FDR correction, there was no significant association between social media addiction and cortical thickness of any ROI.

3.2. Vertexwise analysis of brain morphology

Vertexwise models were performed to analyze cortical morphology at a greater spatial resolution than ROI analysis. FEMA was utilized to run a cross-sectional regression model using vertexwise measures of cortical thickness and surface area versus average daily social media use or addiction. Z-scored beta coefficients were projected onto a three-dimensional heat map for daily social media use (Fig. 2A, 2C) and social media addiction (Fig. 2B, 2D). Visualizations were constructed with a cutoff of a raw p -value of .05, and unthresholded heatmaps and versions with a stricter cutoff of $p < .01$ are shown in Supplemental Figure A.2.

As illustrated in Fig. 2A, greater average daily social media use was associated with areas of lower cortical thickness across a broad range of cortical regions across all lobes. Cortical regions with significant associations were consistent with those identified in ROI analysis. Regions with strong effects were located along the precuneus, inferior temporal/fusiform, superior temporal, supra-marginal, inferior parietal, dorso-lateral prefrontal, and anterior cingulate regions, and these regions

Table 2

Associations between social media use and addiction with region-of-interest (ROI) cortical thickness.

(a) Q-values are Benjamini-Hochberg FDR adjusted p -value	Social media time				Social media addiction			
	Left Hemisphere		Right Hemisphere		Left Hemisphere		Right Hemisphere	
	x100 β , mean z-scored thickness, $\pm$ SEM	p /Q-values (a)	x100 β , mean z-scored thickness, $\pm$ SEM	p /Q-values (a)	x100 β , mean z-scored thickness, $\pm$ SEM	p /Q-values (a)	x100 β , mean z-scored thickness, $\pm$ SEM	p /Q-values (a)
Frontal								
Superior frontal	-2.3 $\pm$ 1.2	4.6e-2/7.4e-2	-3.0 $\pm$ 1.2	9.3e-3/2.5e-2	1.3 $\pm$ 1.2	2.8e-1/9.7e-1	0.4 $\pm$ 1.2	7.6e-1/9.7e-1
Caudal middle frontal	-2.9 $\pm$ 1.2	1.2e-2/2.8e-2	-4.1 $\pm$ 1.2	5.7e-4/9.7e-3	-0.1 $\pm$ 1.2	9.6e-1/9.8e-1	-0.7 $\pm$ 1.2	5.7e-1/9.7e-1
Rostral middle frontal	-2.0 $\pm$ 1.1	7.8e-2/1.1e-1	-2.9 $\pm$ 1.2	1.4e-2/3.0e-2	0.1 $\pm$ 1.1	9.0e-1/9.8e-1	0.9 $\pm$ 1.2	4.3e-1/9.7e-1
Pars opercularis	-1.7 $\pm$ 1.2	1.6e-1/2.0e-1	-0.5 $\pm$ 1.2	6.9e-1/7.2e-1	0.3 $\pm$ 1.2	8.0e-1/9.7e-1	0.5 $\pm$ 1.2	6.9e-1/9.7e-1
Pars triangularis	-2.3 $\pm$ 1.2	5.6e-2/8.5e-2	-3.1 $\pm$ 1.2	1.0e-2/2.6e-2	0.8 $\pm$ 1.2	4.9e-1/9.7e-1	0.2 $\pm$ 1.2	9.0e-1/9.8e-1
Pars orbitalis	-2.6 $\pm$ 1.2	3.3e-2/5.9e-2	-3.1 $\pm$ 1.2	9.3e-3/2.5e-2	2.3 $\pm$ 1.2	6.3e-2/9.7e-1	-0.4 $\pm$ 1.2	7.7e-1/9.7e-1
Lateral orbitofrontal	-2.9 $\pm$ 1.2	1.4e-2/3.0e-2	-3.4 $\pm$ 1.2	3.3e-3/1.8e-2	0.4 $\pm$ 1.2	7.3e-1/9.7e-1	0.1 $\pm$ 1.2	9.3e-1/9.8e-1
Medial orbitofrontal	-2.5 $\pm$ 1.2	3.4e-2/5.9e-2	-2.9 $\pm$ 1.2	1.5e-2/3.2e-2	-0.6 $\pm$ 1.2	5.9e-1/9.7e-1	0.2 $\pm$ 1.2	8.7e-1/9.8e-1
Paracentral	-2.1 $\pm$ 1.2	7.0e-2/1.0e-1	-3.0 $\pm$ 1.2	1.1e-2/2.7e-2	0.3 $\pm$ 1.2	7.7e-1/9.7e-1	0.8 $\pm$ 1.2	4.8e-1/9.7e-1
Precentral	-2.0 $\pm$ 1.2	7.6e-2/1.1e-1	-2.0 $\pm$ 1.2	9.9e-2/1.3e-1	0.3 $\pm$ 1.2	8.1e-1/9.7e-1	0.5 $\pm$ 1.2	6.6e-1/9.7e-1
Frontal pole	-0.4 $\pm$ 1.2	7.5e-1/7.7e-1	-2.9 $\pm$ 1.2	1.8e-2/3.5e-2	-0.4 $\pm$ 1.2	7.7e-1/9.7e-1	-0.5 $\pm$ 1.2	6.8e-1/9.7e-1
Parietal								
Superior parietal	-3.2 $\pm$ 1.1	5.0e-3/2.1e-2	-3.8 $\pm$ 1.2	9.2e-4/1.1e-2	-1.0 $\pm$ 1.2	4.0e-1/9.7e-1	-0.0 $\pm$ 1.2	9.7e-1/9.8e-1
Inferior parietal	-4.8 $\pm$ 1.2	3.1e-5/1.0e-3	-3.6 $\pm$ 1.1	1.6e-3/1.2e-2	-1.0 $\pm$ 1.2	3.8e-1/9.7e-1	-0.8 $\pm$ 1.1	4.9e-1/9.7e-1
Supramarginal	-3.8 $\pm$ 1.2	1.2e-3/1.1e-2	-3.1 $\pm$ 1.1	5.9e-3/2.2e-2	-0.6 $\pm$ 1.2	6.0e-1/9.7e-1	-0.0 $\pm$ 1.1	9.9e-1/9.9e-1
Postcentral	-2.6 $\pm$ 1.1	1.9e-2/3.5e-2	-3.0 $\pm$ 1.1	8.1e-3/2.4e-2	-0.3 $\pm$ 1.1	7.9e-1/9.7e-1	-0.5 $\pm$ 1.1	6.4e-1/9.7e-1
Precuneus	-3.5 $\pm$ 1.2	2.5e-3/1.6e-2	-3.8 $\pm$ 1.2	1.0e-3/1.1e-2	-0.3 $\pm$ 1.2	7.9e-1/9.7e-1	-0.4 $\pm$ 1.2	7.4e-1/9.7e-1
Temporal								
Superior temporal	-2.3 $\pm$ 1.1	4.6e-2/7.4e-2	-2.1 $\pm$ 1.2	7.6e-2/1.1e-1	0.5 $\pm$ 1.2	6.6e-1/9.7e-1	0.3 $\pm$ 1.2	8.1e-1/9.7e-1
Middle temporal	-1.2 $\pm$ 1.1	3.1e-1/3.5e-1	-2.0 $\pm$ 1.1	8.2e-2/1.1e-1	-0.4 $\pm$ 1.1	7.4e-1/9.7e-1	-0.6 $\pm$ 1.1	6.0e-1/9.7e-1
Inferior temporal	-3.4 $\pm$ 1.2	4.3e-3/1.9e-2	-3.4 $\pm$ 1.2	3.4e-3/1.8e-2	-2.3 $\pm$ 1.2	5.3e-2/9.7e-1	0.5 $\pm$ 1.2	6.9e-1/9.7e-1
Banks of the superior temporal sulcus	-5.1 $\pm$ 1.2	1.8e-5/1.0e-3	-3.2 $\pm$ 1.2	8.2e-3/2.4e-2	-1.4 $\pm$ 1.2	2.5e-1/9.7e-1	-2.3 $\pm$ 1.2	6.1e-2/9.7e-1
Fusiform	-2.9 $\pm$ 1.2	1.3e-2/3.0e-2	-2.9 $\pm$ 1.1	1.0e-2/2.6e-2	0.4 $\pm$ 1.2	7.2e-1/9.7e-1	0.9 $\pm$ 1.1	4.5e-1/9.7e-1
Transverse temporal	-3.8 $\pm$ 1.2	1.3e-3/1.1e-2	-2.4 $\pm$ 1.2	3.8e-2/6.4e-2	0.4 $\pm$ 1.2	7.1e-1/9.7e-1	1.4 $\pm$ 1.2	2.2e-1/9.7e-1
Entorhinal	-1.5 $\pm$ 1.2	2.1e-1/2.5e-1	-1.2 $\pm$ 1.2	3.1e-1/3.5e-1	-0.5 $\pm$ 1.2	6.7e-1/9.7e-1	0.2 $\pm$ 1.2	8.5e-1/9.8e-1
Temporal pole	-1.2 $\pm$ 1.2	3.3e-1/3.7e-1	-1.4 $\pm$ 1.2	2.5e-1/3.0e-1	0.1 $\pm$ 1.2	9.1e-1/9.8e-1	-0.1 $\pm$ 1.2	9.5e-1/9.8e-1
Parahippocampal	-1.3 $\pm$ 1.2	2.6e-1/3.1e-1	-0.9 $\pm$ 1.2	4.5e-1/4.8e-1	-0.3 $\pm$ 1.2	7.8e-1/9.7e-1	0.3 $\pm$ 1.2	8.0e-1/9.7e-1
Occipital								
Lateral occipital	-2.9 $\pm$ 1.0	4.0e-3/1.9e-2	-3.7 $\pm$ 1.0	2.5e-4/5.7e-3	-0.9 $\pm$ 1.0	3.9e-1/9.7e-1	-1.2 $\pm$ 1.0	2.3e-1/9.7e-1
Lingual	-2.9 $\pm$ 1.1	8.0e-3/2.4e-2	-3.0 $\pm$ 1.1	8.2e-3/2.4e-2	-1.1 $\pm$ 1.1	3.2e-1/9.7e-1	-0.3 $\pm$ 1.1	7.8e-1/9.7e-1
Cuneus	-2.6 $\pm$ 1.2	2.3e-2/4.2e-2	-3.0 $\pm$ 1.1	8.0e-3/2.4e-2	-1.2 $\pm$ 1.2	2.9e-1/9.7e-1	-1.5 $\pm$ 1.1	1.8e-1/9.7e-1
Pericalcarine	-2.2 $\pm$ 1.1	4.8e-2/7.6e-2	-3.4 $\pm$ 1.1	2.6e-3/1.6e-2	-1.0 $\pm$ 1.1	3.6e-1/9.7e-1	-1.1 $\pm$ 1.1	3.3e-1/9.7e-1
Other								
Rostral anterior cingulate	-3.2 $\pm$ 1.2	5.7e-3/2.2e-2	-0.0 $\pm$ 1.2	9.7e-1/9.7e-1	-0.4 $\pm$ 1.2	7.1e-1/9.7e-1	-0.3 $\pm$ 1.2	8.0e-1/9.7e-1
Caudal anterior cingulate	-1.3 $\pm$ 1.2	2.8e-1/3.3e-1	-1.7 $\pm$ 1.2	1.6e-1/2.0e-1	-1.3 $\pm$ 1.2	2.9e-1/9.7e-1	-0.9 $\pm$ 1.2	4.6e-1/9.7e-1
Posterior cingulate	-0.8 $\pm$ 1.2	5.2e-1/5.5e-1	-1.2 $\pm$ 1.2	3.3e-1/3.7e-1	1.4 $\pm$ 1.2	2.3e-1/9.7e-1	1.4 $\pm$ 1.2	2.5e-1/9.7e-1
Isthmus cingulate	0.1 $\pm$ 1.2	9.4e-1/9.6e-1	-2.0 $\pm$ 1.2	8.9e-2/1.2e-1	1.2 $\pm$ 1.2	3.2e-1/9.7e-1	-1.0 $\pm$ 1.2	4.3e-1/9.7e-1
Insula	-2.2 $\pm$ 1.2	5.5e-2/8.5e-2	-2.7 $\pm$ 1.2	1.8e-2/3.5e-2	0.1 $\pm$ 1.2	9.6e-1/9.8e-1	-0.4 $\pm$ 1.2	7.3e-1/9.7e-1
All Cortical Gray Matter	-4.1 $\pm$ 1.1	2.7e-4	-4.6 $\pm$ 1.1	4.9e-5	-0.3 $\pm$ 1.1	8.0e-1	-0.1 $\pm$ 1.1	9.0e-1

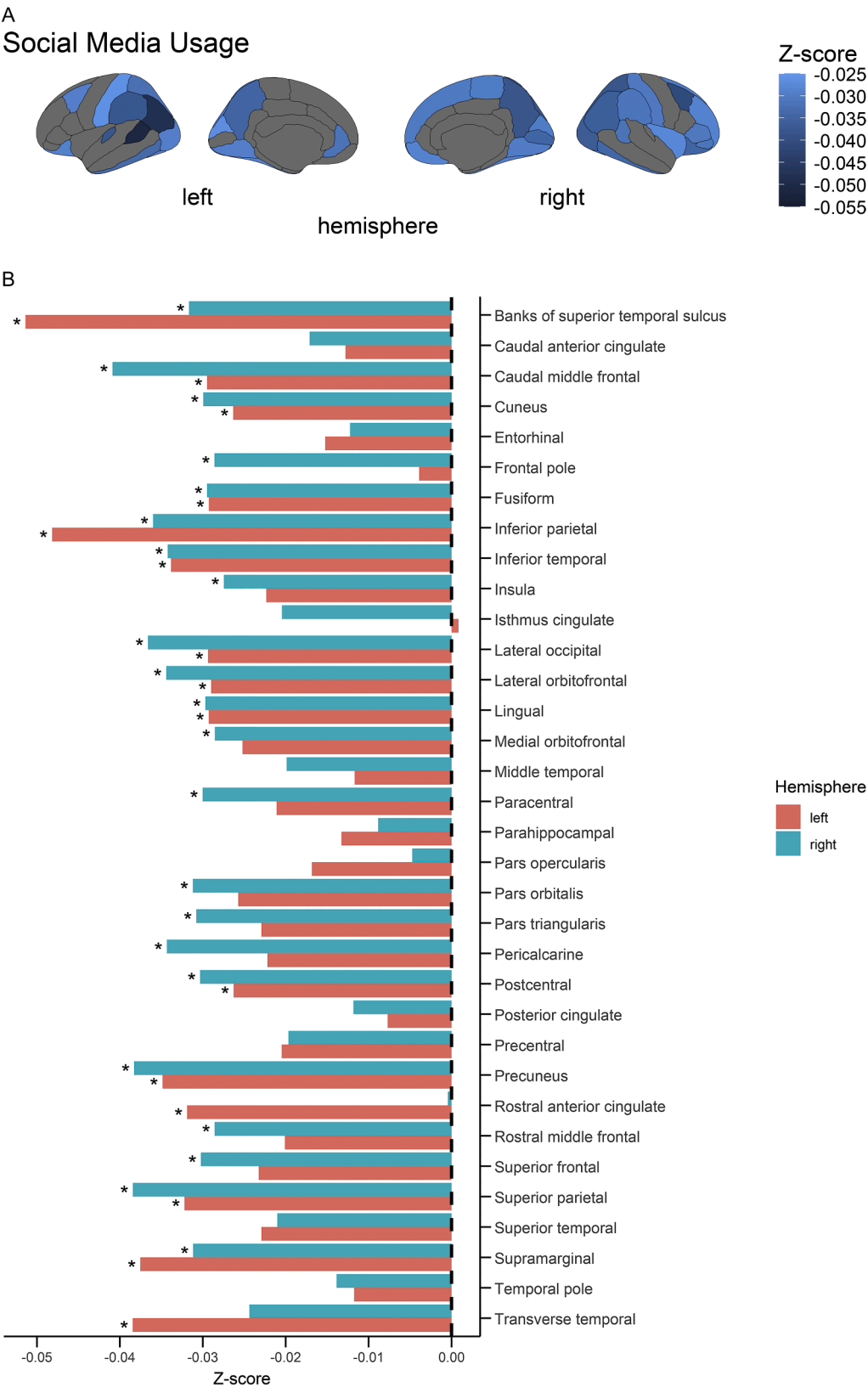
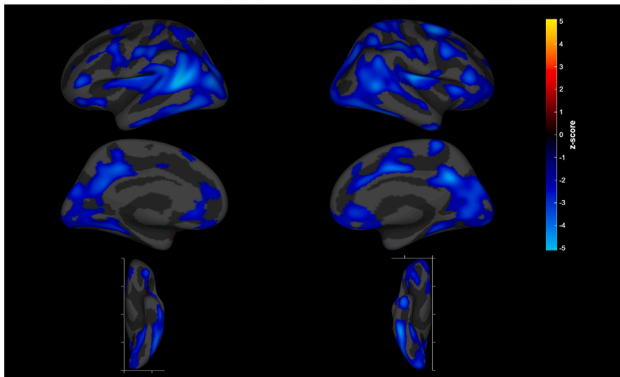
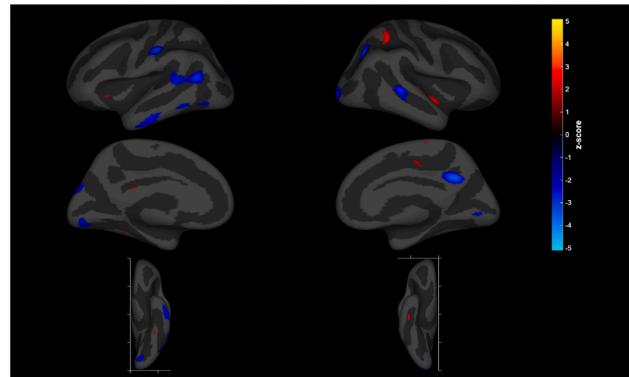


Fig. 1. Visualization of associations between average daily social media use and cortical ROI thickness. (A) shows standardized beta coefficients projected onto a two-dimensional map of cortical ROIs. Only regions with $Q < 0.05$ are shown. (B) shows a bar graph of standardized beta coefficients for the association between social media use with left and right hemisphere ROIs. *denotes significance at $Q < 0.05$.

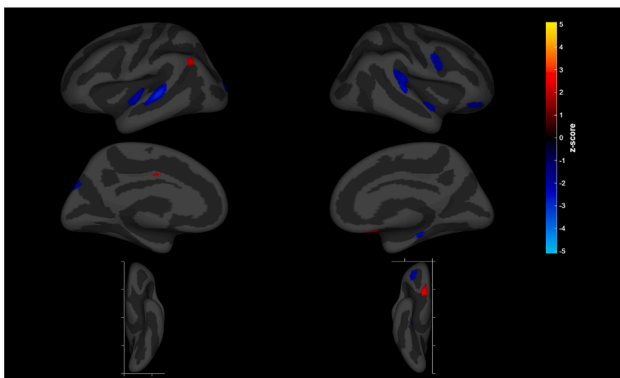
A. Cortical thickness vs. social media usage



B. Cortical thickness vs. social media addiction



C. Cortical surface area vs. social media usage



D. Cortical surface area vs. social media addiction

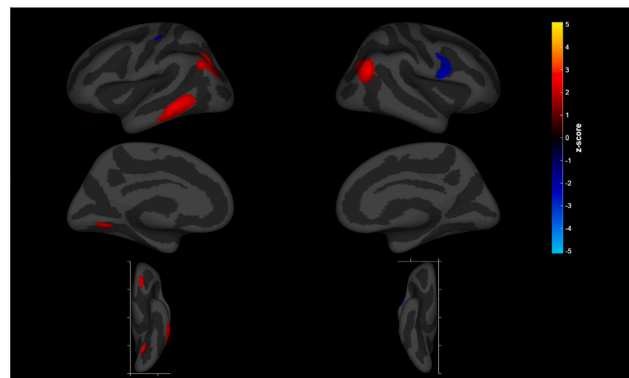


Fig. 2. Surface projections of vertexwise analysis on inflated cortical surfaces. At each vertex, the heat maps show beta coefficient Z-scores for (A) social media use with cortical thickness, (B) social media addiction with cortical thickness, (C) social media usage with surface area, and (D) social media addiction with surface area from multivariate linear mixed effects models, adjusted for demographics, family socioeconomic status, genetic ancestry, non-social media screen time, MRI scanner serial number, and MRI scanner software version. Z-scores are thresholded at a raw p-value of 0.05. Views of the brain are left lateral (top row, left), right lateral (top row, right), left medial (middle row, left), right medial (middle row, right), left ventral (bottom row, left), and right ventral (bottom row, right).

survived stricter thresholding. In addition, there were small regions in the bilateral superior temporal regions with decreased surface area that survived stricter thresholding (Supplemental Figure A.2C).

For social media addiction, there were small cortical regions in the left and right parietal lobes, left and right temporal lobes, right occipital lobe, and right precuneus with lower thickness and a small region of higher thickness in the superior temporal lobe which survived stricter thresholding (Supplemental Figure A.2B). There were also small regions in the right and left parietal lobes with higher surface area associated with social media addiction (Supplemental Figure A.2D).

Previous research has found that ADHD may be associated with media use (Beyens et al., 2018; Rauschecker et al., 2025), thus we also performed vertexwise analysis with the Child Behavior Checklist ADHD t-score as an additional co-regressor (Supplemental Figure A.3). There were no appreciable changes in the spatial distribution or magnitude of the associations.

4. Discussion

Adolescent social media use and addiction have become increasingly prevalent in recent years, raising concerns among caregivers and providers. However, the relationship between these behaviors and structural brain development remains underexplored. In a demographically diverse sample of over 7,000 U.S. adolescents, we found that greater daily social media use was associated with reduced cortical thickness in a wide array of cortical regions that remained significant after FDR

correction and adjustment for age, sex, race, genetic ancestry, caregiver partnership, household income, highest caregiver education, non-social media screen time, and MRI scanner serial number and software version. Greater daily social media use was not associated with greater cortical thickness in any brain region. No significant associations were observed between social media addiction and cortical morphology.

The potential associations between social media use and brain structure are especially important during early adolescence, when social media use typically escalates rapidly, and the brain undergoes rapid neurodevelopmental changes. In the developing brain, typical neurodevelopment includes gradual reductions in cortical thickness (Squeglia et al., 2013) across the cerebral cortex in a regionally and temporally organized manner (Ducharme et al., 2016). This reduction in cortical thickness is hypothesized to primarily reflect synaptic pruning, which supports the development of functionally distributed neural networks (Fair et al., 2009), and increased myelination of white matter (Marsh et al., 2008; Tamnes et al., 2017). However, excessive reductions in cortical thickness during adolescence may be a marker of abnormal brain development and have been associated with deficits in emotional regulation (Albaugh et al., 2023; Ewell et al., 2023; Xiao et al., 2023), deficits in cognitive functioning (Shaw et al., 2006), and the development of psychiatric disorders (Schmaal et al., 2017).

Cortical thickness, surface area, and volume are influenced by distinct genetic factors (Panizzon et al., 2009; Winkler et al., 2010) and follow largely independent developmental trajectories. In contrast to cortical thickness and volume, which typically decline throughout

adolescence, cortical surface area increases throughout early childhood and typically peaks around age 10 before declining (Wierenga et al., 2014). Regional brain maturation occurs at different rates, and areas involved in shared functional networks often develop in tandem (Alexander-Bloch et al., 2013; Geng et al., 2017; Sotiras et al., 2017). Cortical surface area appears to be more strongly influenced by genetic factors and is associated with more genetic loci (Grasby et al., 2020), while cortical thickness appears to be more susceptible to environmental influence (Panizzon et al., 2009; van Drunen et al., 2024). These discrepant bases of cortical thickness versus cortical surface area may offer important insights into our findings illustrating associations of lower cortical thickness in widespread cortical regions with greater social media use, while cortical surface area was relatively undifferentiated.

Assessment of the regional patterns of lower cortical thickness associated with social media use illustrated significant differences in the precuneus, lateral parietal cortex, medial and inferior prefrontal cortices, inferior temporal gyrus, and lateral occipital cortex—regions that overlap with key nodes of the default mode network (DMN), prefrontal executive control networks, and visual processing and attention networks (Goodale and Milner, 1992). Interestingly, we did not observe differences in the volume of subcortical structures, although microstructural variations in these regions may occur in the absence of macrostructural change.

The cortical regions involved in visual processing that showed lower thickness in relation to greater social media use in our study have also been implicated in various forms of psychopathology. Children with dyslexia have exhibited decreased cortical thickness in the left inferior parietal, lateral occipital, inferior temporal, and fusiform areas (Williams et al., 2018). Decreased thickness in the fusiform gyrus has been observed in children with ADHD (Hoogman et al., 2019), but relationships between social media use and cortical morphology persisted after adjusting for ADHD score.

We also found significant associations between social media use and cortical structure in frontal executive regions, particularly the superior frontal gyrus and dorsolateral prefrontal cortex/caudal middle frontal gyrus. These regions are involved in higher-order cognitive functions, such as cognitive control (Hare et al., 2009; Miller and Cohen, 2001), planning, strategic reasoning (Yamagishi et al., 2016), and emotional and conflict management (Kuehne et al., 2019; Oehrn et al., 2014). Frontal regions in the right hemisphere, including portions of the middle frontal gyrus, have been associated with attentional reorienting and proposed to support interactions between dorsal and ventral attention systems (Corbetta et al., 2008; Japee et al., 2015). The functional connectivity of the right middle frontal gyrus (rMFG) with ventral and dorsal attention networks may be implicated in ADHD (Epstein et al., 2009; Wang et al., 2025). Higher cortical thickness in the rMFG has been associated with lower impulsivity in adolescents, independent of age (Pehlivanova et al., 2018). Another study showed higher thickness in the rMFG among adolescents was associated with improved planning skills (Kollndorfer et al., 2023), but this study was limited by a small sample size with participants spanning a wide age range. Additionally, functional connectivity in the left dorsolateral prefrontal cortex is also implicated in major depressive disorder and is the target of transcranial magnetic stimulation (Fox et al., 2012). We also observed lower thickness in the left rostral anterior cingulate, a region involved in emotion regulation (Stevens et al., 2011), with similar patterns observed in adolescents with substance use disorders (Boulos et al., 2016).

Lower cortical thickness was also observed in association with greater social media use in nodes of the DMN—the precuneus, inferior parietal lobule, supramarginal gyrus, and dorsomedial prefrontal cortex (Raichle, 2015). The DMN is classically associated with self-referential thought and resting-state brain activity (Andrews-Hanna et al., 2010; Kucyi et al., 2016), though its full role remains debated. Prior research has linked the thickness of DMN structures to general intelligence in young adults (Yadav and Purushotham, 2025), and disruptions to DMN

development have been linked to adolescent psychopathology (Broulidakis et al., 2016; Jirsaraie et al., 2025; Lee et al., 2024; Padmanabhan et al., 2017; Xiao et al., 2023; Yan et al., 2019).

4.1. Strengths and limitations

This study is subject to several limitations. The reliance on self-reported social media use data introduces the potential for biases, such as inaccurate reporting or social desirability bias. Although adolescents may under-report their social media use, prior work in the ABCD Study suggests that parents tend to report lower levels of adolescents' social media use than adolescents themselves (Nagata et al., 2022). Adolescent self-report may therefore capture aspects of social media use that are not readily observable by parents, such as underage use or brief private use, although multimethod approaches remain an important direction for future research. Moreover, while our findings suggest an association between social media use and lower cortical thickness, the cross-sectional nature of the data limits any claims of causality or directionality. Although we adjusted for several known confounders, including demographics, socioeconomic factors, genetic ancestry, and scanner features, residual confounding may still exist. It is also possible that excessive social media use may be a marker for pre-existing neurobiological differences rather than a causal factor. As the ABCD Study progresses, the longitudinal effects of social media can be further studied with causal inference analyses. Despite these limitations, each factor can independently account for unique variations of cortical anatomy, highlighting non-overlapping effects.

It is important to note that this study focuses exclusively on one dimension of adolescent life—social media use—operationalized through general usage and self-reported addiction metrics. This singular focus may not capture the full range of relevant environmental and behavioral factors that may also be associated with neuroanatomical outcomes. For instance, parental monitoring may influence the extent of social media use in adolescents, introducing variability not accounted for in this model. Furthermore, we did not assess the nature or content of social media engagement, which varies widely among users and may play a role in shaping neurodevelopment.

A key strength of this study is the use of a large, demographically and geographically diverse population of participants. Another strength is the use of vertexwise analysis, which provides high spatial resolution and an unbiased view of cortical morphology compared with ROI analysis.

Although this study identified negative associations between cortical thickness and social media exposure within regions relevant to cognitive functions and psychopathology, we caution against drawing conclusions about the harmful effects of social media use. We did not find any significant associations between cortical morphology and social media addiction. This may be due to the lower levels of social media use at this time point in the early adolescent sample, and the potential effects of social media addiction may not have had time to accumulate, if they do exist. Future research studying functional measures and adolescents at different points in development is needed.

It is also worth noting that the effect sizes of social media use, while statistically significant, were relatively small; significant standardized beta coefficients for the association between ROI thickness and social media usage ranged from -0.051 to -0.026. The magnitude of these associations was comparable to that of television viewing and reading in a previous study (Rauschecker et al., 2025).

Although factors like genetics and socioeconomic status may have stronger associations with brain morphology, social media use is worth studying because it is a modifiable risk factor. Due to the cross-sectional design, the causality and directionality of the observed associations cannot be inferred. Longitudinal studies incorporating both neuroimaging and behavioral assessments will be needed to further elucidate the effects of social media use on adolescent neurodevelopment.

5. Ethics approval and consent to participate

The University of California, San Diego (UCSD) provided centralized institutional review board (IRB) approval and each participating site received local IRB approval:

- Children's Hospital Los Angeles, Los Angeles, California.
- Florida International University, Miami, Florida.
- Laureate Institute for Brain Research, Tulsa, Oklahoma.
- Medical University of South Carolina, Charleston, South Carolina.
- Oregon Health and Science University, Portland, Oregon.
- SRI International, Menlo Park, California.
- University of California San Diego, San Diego, California.
- University of California Los Angeles, Los Angeles, California.
- University of Colorado Boulder, Boulder, Colorado.
- University of Florida, Gainesville, Florida.
- University of Maryland at Baltimore, Baltimore, Maryland.
- University of Michigan, Ann Arbor, Michigan.
- University of Minnesota, Minneapolis, Minnesota.
- University of Pittsburgh, Pittsburgh, Pennsylvania.
- University of Rochester, Rochester, New York.
- University of Utah, Salt Lake City, Utah.
- University of Vermont, Burlington, Vermont.
- University of Wisconsin—Milwaukee, Milwaukee, Wisconsin.
- Virginia Commonwealth University, Richmond, Virginia.
- Washington University in St. Louis, St. Louis, Missouri.
- Yale University, New Haven, Connecticut.

Written informed consent was obtained from the parents/caregivers of adolescents, and written assent was obtained from adolescents. Given that adolescent participants were minors (10–14 years old), they were not able to give legal consent. The study adhered to the Declaration of Helsinki.

Funding sources

The research was supported by the National Institutes of Health (R01DA064134, K08HL159350 and R01MH135492). The funders had no role in the study analysis, decision to publish the study, or the preparation of the manuscript. The ABCD Study was supported by the National Institutes of Health and additional federal partners under award numbers U01DA041022, U01DA041025, U01DA041028, U01DA041048, U01DA041089, U01DA041093, U01DA041106, U01DA041117, U01DA041120, U01DA041134, U01DA041148, U01DA041156, U01DA041174, U24DA041123, and U24DA041147. A full list of supporters is available at <https://abcdstudy.org/federal-partners/>. A listing of participating sites and a complete listing of the study investigators can be found at <https://abcdstudy.org/principal-investigators.html>. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in analysis or writing of this report. The statements in this article are solely the responsibility of the authors and do not necessarily represent the views of the National Institutes of Health.

Availability of data and materials

Data used in the preparation of this article were obtained from the ABCD Study (<https://abcdstudy.org>), held in the NIH Brain Development Cohorts (NBDC) Portal.

CRediT authorship contribution statement

Jason M. Nagata: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Kevin Bao:** Writing – review & editing, Writing – original draft, Formal analysis. **Stuart B. Murray:** Writing – review & editing, Writing – original draft.

Pierre Nedelec: Writing – review & editing, Writing – original draft. **Racquel A. Richardson:** Writing – review & editing, Writing – original draft. **Sahana Nayak:** Writing – review & editing, Writing – original draft. **Elizabeth J. Li:** Writing – review & editing, Writing – original draft. **Jennifer H. Wong:** Writing – review & editing, Writing – original draft. **Eva M. Muller-Oehring:** Writing – review & editing. **Aaron Scheffler:** Writing – review & editing. **Fiona C. Baker:** Writing – review & editing, Data curation, Conceptualization. **Andreas M. Rauschecker:** Writing – review & editing. **Leo P. Sugrue:** Writing – review & editing.

Declaration of competing interest

Dr. Murray has testified as an expert witness in litigation involving social media companies. The other authors have no conflicts to declare.

Acknowledgements

We thank Anthony Kung for editorial assistance.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.neuroimage.2026.121860](https://doi.org/10.1016/j.neuroimage.2026.121860).

References

- Achterberg, M., Becht, A., van der Cruysen, R., van de Groep, I.H., Spaans, J.P., Klapwijk, E., Crone, E.A., 2022. Longitudinal associations between social media use, mental well-being and structural brain development across adolescence. *Dev. Cogn. Neurosci.* 54, 101088. <https://doi.org/10.1016/j.dcn.2022.101088>.
- Albaugh, M.D., Hudziak, James.J., Spechler, P.A., Chaarani, B., Lepage, C., Jeon, S., Rioux, P., Evans, A.C., Banaschewski, T., Bokde, A.L.W., Desrivieres, S., Flor, H., Gowland, P., Heinz, A., Ittermann, B., Martinot, J.-L., Martinot, M.-L.P., Nees, F., Orfanos, D.P., Poustka, L., Millenet, S., Fröhner, J.H., Smolka, M.N., Walter, H., Whelan, R., Schumann, G., Potter, A.S., Garavan, H., 2023. Conduct problems are associated with accelerated thinning of emotion-related cortical regions in a community-based sample of adolescents. *Psychiatry Res.* 330, 111614. <https://doi.org/10.1016/j.psychres.2023.111614>.
- Alexander-Bloch, A., Raznahan, A., Bullmore, E., Giedd, J., 2013. The convergence of maturational change and structural covariance in human cortical networks. *J. Neurosci.* 33, 2889–2899. <https://doi.org/10.1523/JNEUROSCI.3554-12.2013>.
- Andreassen, C.S., Torbjørn, T., Brunborg, G.S., Pallesen, S., 2012. Development of a facebook addiction scale. *Psychol. Rep.* 110, 501–517. <https://doi.org/10.2466/02.09.18.PR0.110.2.501-517>.
- Andrews-Hanna, J.R., Reidler, J.S., Huang, C., Buckner, R.L., 2010. Evidence for the default network's role in spontaneous cognition. *J. Neurophysiol.* 104, 322–335. <https://doi.org/10.1152/jn.00830.2009>.
- Araín, M., Haque, M., Johal, L., Mathur, P., Nel, W., Rais, A., Sandhu, R., Sharma, S., 2013. Maturation of the adolescent brain. *Neuropsychiatr. Dis. Treat.* 9, 449–461. <https://doi.org/10.2147/NDT.S39776>.
- Bagot, K., Tomko, R., Marshall, A.T., Hermann, J., Cummins, K., Ksian, A., Kakalis, M., Breslin, F., Lisdahl, K., Mason, M., Redhead, J., Squeglia, L., Thompson, W., Wade, T., Tapert, S., Fuemmeler, B., Baker, F., 2022. Youth screen use in the ABCD® study. *Dev. Cogn. Neurosci.* 57, 101150. <https://doi.org/10.1016/j.dcn.2022.101150>.
- Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. B (Methodol.)* 57, 289–300.
- Bernanke, J., Luna, A., Chang, L., Bruno, E., Dworkin, J., Posner, J., 2022. Structural brain measures among children with and without ADHD in the Adolescent Brain and Cognitive Development Study cohort: a cross-sectional US population-based study. *Lancet Psychiat.* 9, 222–231. [https://doi.org/10.1016/S2215-0366\(21\)00505-8](https://doi.org/10.1016/S2215-0366(21)00505-8).
- Beyens, I., Valkenburg, P.M., Piotrowski, J.T., 2018. Screen media use and ADHD-related behaviors: four decades of research. *Proc. Natl. Acad. Sci.* 115, 9875–9881. <https://doi.org/10.1073/pnas.1611611114>.
- Boulos, P.K., Dalwani, M.S., Tanabe, J., Mikulich-Gilbertson, S.K., Banich, M.T., Crowley, T.J., Sakai, J.T., 2016. Brain cortical thickness differences in adolescent females with substance use disorders. *Plos One.* 11, e0152983. <https://doi.org/10.1371/journal.pone.0152983>.
- Broulidakis, M.J., Fairchild, G., Sully, K., Blumensath, T., Darekar, A., Sonuga-Barke, E.J.S., 2016. Reduced default mode connectivity in adolescents with conduct disorder. *J. Am. Acad. Child Adolesc. Psychiat.* 55, e1. <https://doi.org/10.1016/j.jaac.2016.05.021>.
- Casey, B.J., Cannonier, T., Conley, M.I., Cohen, A.O., Barch, D.M., Heitzeg, M.M., Soules, M.E., Teslovich, T., Dellarco, D.V., Garavan, H., Orr, C.A., Wager, T.D., Banich, M.T., Speer, N.K., Sutherland, M.T., Riedel, M.C., Dick, A.S., Bjork, J.M., Thomas, K.M., Chaarani, B., Mejia, M.H., Hagler, D.J., Daniela Cornejo, M., Scat, C.S., Harms, M.P., Dosenbach, N.U.F., Rosenberg, M., Earl, E., Bartsch, H., Watts, R., Polimeni, J.R., Kuperman, J.M., Fair, D.A., Dale, A.M., ABCD Imaging Acquisition

- Workgroup, 2018. The Adolescent Brain Cognitive Development (ABCD) study: imaging acquisition across 21 sites. *Dev. Cogn. Neurosci.* 32, 43–54. <https://doi.org/10.1016/j.dcn.2018.03.001>.
- Chen, Y.-Y., Yim, H., Lee, T.-H., 2023. Negative impact of daily screen use on inhibitory control network in preadolescence: a two-year follow-up study. *Dev. Cogn. Neurosci.* 60, 101218. <https://doi.org/10.1016/j.dcn.2023.101218>.
- Corbetta, M., Patel, G., Shulman, G.L., 2008. The reorienting system of the human brain: from environment to theory of mind. *Neuron* 58, 306–324. <https://doi.org/10.1016/j.neuron.2008.04.017>.
- Deng, H., Song, K., Geng, X., Xu, L., Zhang, Jialin, Li, X., He, J., Potenza, M.N., Zhang, Jintao, 2025. Online social activity time predicts ADHD problems in youth from late childhood to early adolescence in the ABCD study. *Eur. Child Adolesc. Psychiat.* 34, 2195–2204. <https://doi.org/10.1007/s00787-024-02620-6>.
- Desikan, R.S., Ségonne, F., Fischl, B., Quinn, B.T., Dickerson, B.C., Blacker, D., Buckner, R.L., Dale, A.M., Maguire, R.P., Hyman, B.T., Albert, M.S., Killiany, R.J., 2006. An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *NeuroImage* 31, 968–980. <https://doi.org/10.1016/j.neuroimage.2006.01.021>.
- Ducharme, S., Albaugh, M.D., Nguyen, T.-V., Hudziak, J.J., Mateos-Pérez, J.M., Labbe, A., Evans, A.C., Karama, S., 2016. Trajectories of cortical thickness maturation in normal brain development – the importance of quality control procedures. *Neuroimage* 125, 267–279. <https://doi.org/10.1016/j.neuroimage.2015.10.010>.
- Epstein, J.N., Delbello, M.P., Adler, C.M., Altaye, M., Kramer, M., Mills, N.P., Strakowski, S.M., Holland, S., 2009. Differential patterns of brain activation over time in adolescents with and without attention deficit hyperactivity disorder (ADHD) during performance of a sustained attention task. *Neuropsychiatry* 40, 1–5. <https://doi.org/10.1055/s-0029-1220686>.
- Ewell, A., Allard, T., Botdorf, M., Ji, A., Riggins, T., 2023. Emotion regulation and reactivity are associated with cortical thickness in early to mid-childhood. *Dev. Psychobiol.* 65, e22412. <https://doi.org/10.1002/dev.22412>.
- Fair, D.A., Cohen, A.L., Power, J.D., Dosenbach, N.U.F., Church, J.A., Miezin, F.M., Schlaggar, B.L., Petersen, S.E., 2009. Functional brain networks develop from a “local to distributed” organization. *Plos Comput. Biol.* 5, e1000381. <https://doi.org/10.1371/journal.pcbi.1000381>.
- Fan, C.C., Bartsch, H., Schork, A., Chen, C.-H., Wang, Y., Lo, M.-T., Brown, T.T., Kuperman, J.M., Hagler, D.J., Schork, N., Jernigan, T.L., Dale, A.M., 2015. Modeling the 3D geometry of the cortical surface with genetic ancestry. *Curr. Biol.* 25, 1988–1992. <https://doi.org/10.1016/j.cub.2015.06.006>.
- Fischl, B., Salat, D.H., Busa, E., Albert, M., Dieterich, M., Haselgrove, C., Van der Kouwe, A., Killiany, R., Kennedy, D., Klaveness, S., Montillo, A., Makris, N., Rosen, B., Dale, A.M., 2002. Whole brain segmentation: automated labeling of neuroanatomical structures in the human brain. *Neuron* 33, 341–355. [https://doi.org/10.1016/S0896-6273\(02\)00569-X](https://doi.org/10.1016/S0896-6273(02)00569-X).
- Fox, M.D., Buckner, R.L., White, M.P., Greicius, M.D., Pascual-Leone, A., 2012. Efficacy of transcranial magnetic stimulation targets for depression is related to intrinsic functional connectivity with the subgenual cingulate. *Biol. Psychiat.* 72, 595–603. <https://doi.org/10.1016/j.biopsych.2012.04.028>.
- Fürtjes, A.E., Cole, J.H., Couvy-Duchesne, B., Ritchie, S.J., 2023. A quantified comparison of cortical atlases on the basis of trait morphometricity. *Cortex* 158, 110–126. <https://doi.org/10.1016/j.cortex.2022.11.001>.
- Garavan, H., Bartsch, H., Conway, K., Decastro, A., Goldstein, R.Z., Heeringa, S., Jernigan, T., Potter, A., Thompson, W., Zahs, D., 2018. Recruiting the ABCD sample: design considerations and procedures. *Dev. Cogn. Neurosci.* 32, 16–22. <https://doi.org/10.1016/j.dcn.2018.04.004>.
- Geng, X., Li, G., Lu, Z., Gao, W., Wang, L., Shen, D., Zhu, H., Gilmore, J.H., 2017. Structural and maturational covariance in early childhood brain development. *Cereb. Cortex* 27, 1795–1807. <https://doi.org/10.1093/cercor/bhw022>.
- Gogarten, S.M., Sofer, T., Chen, H., Yu, C., Brody, J.A., Thornton, T.A., Rice, K.M., Conomos, M.P., 2019. Genetic association testing using the GENESIS R/Bioconductor package. *Bioinformatics* 35, 5346–5348. <https://doi.org/10.1093/bioinformatics/btz567>.
- Goodale, M.A., Milner, A.D., 1992. Separate visual pathways for perception and action. *Trends Neurosci.* 15, 20–25. [https://doi.org/10.1016/0166-2236\(92\)90344-8](https://doi.org/10.1016/0166-2236(92)90344-8).
- Grasby, K.L., Jahanshad, N., Painter, J.N., Colodro-Conde, L., Bralten, J., Hibar, D.P., Lind, P.A., Pizzagalli, F., Ching, C.R.K., McMahon, M.A.B., Shatikhina, Y., Zsembik, L.C.P., Thomopoulos, S.I., Zhu, A.H., Strike, L.T., Agartz, I., Alhusaini, S., Almeida, M.A.A., Alnaes, D., Amlien, I.K., Andersson, M., Ard, T., Armstrong, N.J., Ashley-Koch, A., Atkins, J.R., Bernard, M., Brouwer, R.M., Buimer, E.E.L., Bülow, R., Bürger, C., Cannon, D.M., Chakravarty, M., Chen, Q., Cheung, J.W., Couvy-Duchesne, B., Dale, A.M., Dalvie, S., de Araujo, T.K., de Zubizaray, G.L., de Zwart, S. M.C., den Braber, A., Doan, N.T., Dohm, K., Ehrlich, S., Engelbrecht, H.-R., Erk, S., Fan, C.C., Fedko, I.O., Foley, S.F., Ford, J.M., Fukunaga, M., Garrett, M.E., Ge, T., Giddaluru, S., Goldman, A.L., Green, M.K., Groenewold, N.A., Grotegerd, D., Gurholt, T.P., Gutman, B.A., Hansell, N.J., Harrison, M.A., Harrison, M.B., Haswell, C. C., Hauser, M., Herms, S., Heslenfeld, D.J., Ho, N.F., Hoehn, D., Hoffmann, P., Holleran, L., Hoogman, M., Hottenga, J.-J., Ikeda, M., Janowitz, D., Jansen, I.E., Jia, T., Jockwitz, C., Kanai, R., Karama, S., Kasperaviciute, D., Kaufmann, T., Kelly, S., Kikuchi, M., Klein, M., Knapp, M., Knodt, A.R., Krämer, B., Lam, M., Lancaster, T.M., Lee, P.H., Lett, T.A., Lewis, L.B., Lopes-Cendes, I., Luciano, M., Macciardi, F., Marquand, A.F., Mathias, S.R., Melzer, T.R., Milanese, J., Mirza-Schreiber, N., Moreira, J.C.V., Mühleisen, T.W., Müller-Myhsok, B., Najt, P., Nakahara, S., Nho, K., Olde Loohuis, L.M., Orfanos, D.P., Pearson, J.F., Pitcher, T.L., Pütz, B., Quidé, Y., Ragothaman, A., Rashid, F.M., Reay, W.R., Redlich, R., Reinbold, C.S., Reppe, J., Richard, G., Riedel, B.C., Risacher, S.L., Rocha, C.S., Mota, N.R., Salminen, L., Saremi, A., Saykin, A.J., Schlag, F., Schmaal, L., Schofield, P.R., Secolin, R., Shapland, C.Y., Shen, L., Shin, J., Shumskaya, E., Sønderby, I.E., Sprooten, E., Tansey, K.E., Teumer, A., Thalamuthu, A., Tordesillas-Gutiérrez, D., Turner, J.A., Uhlmann, A., Vallgera, C.L., van der Meer, D., van Donkelaar, M.M.J., van Eijk, L., van Erp, T.G.M., van Haren, N.E.M., van Rooij, D., van Tol, M.-J., Veldink, J.H., Verhoef, E., Walton, E., Wang, M., Wang, Y., Wardlaw, J.M., Wen, W., Westlye, L.T., Whelan, C.D., Witt, S.H., Wittfeld, K., Wolf, C., Wolfers, T., Wu, J.Q., Yasuda, C.L., Zaremba, D., Zhang, Z., Zwiers, M.P., Artiges, E., Assareh, A.A., Ayesa-Arriola, R., Belger, A., Brandt, C.L., Brown, G.G., Cichon, S., Curran, J.E., Davies, G.E., Degenhardt, F., Dennis, M.F., Dietsche, B., Djurovic, S., Doherty, C.P., Espiritu, R., Garijo, D., Gil, Y., Gowland, P.A., Green, R. C., Häusler, A.N., Heindel, W., Ho, B.-C., Hoffmann, W.U., Holsboer, F., Homuth, G., Hosten, N., Jack, C.R., Jang, M., Jansen, A., Kimbrel, N.A., Kolskår, K., Koops, S., Krug, A., Lim, K.O., Luyckx, J.J., Mathalon, D.H., Mather, K.A., Mattay, V.S., Matthews, S., Mayoral Van Son, J., McEwen, S.C., Melle, I., Morris, D.W., Mueller, B. A., Nauck, M., Nordvik, J.E., Nöthen, M.M., O’Leary, D.S., Opel, N., Martinot, M.-L. P., Pike, G.B., Preda, A., Quinlan, E.B., Rasser, P., Ratnakar, V., Reppermund, S., Steen, V.M., Tooney, P.A., Torres, F.R., Veltman, D.J., Voyvodic, J.T., Whelan, R., White, T., Yamamori, H., Adams, H.H.H., Bis, J.C., Dettke, S., Decarli, C., Fornage, M., Gudnason, V., Hofer, E., Ikram, M.A., Launer, L., Longstreth, W.T., Lopez, O.L., Mazoyer, B., Mosley, T.H., Roshchupkin, G.V., Satizabal, C.L., Schmidt, R., Seshadri, S., Yang, Q., Alzheimer’s Disease Neuroimaging Initiative, CHARGE Consortium, EPIGEN Consortium, IMAGEN Consortium, SYS Consortium, Parkinson’s Progression Markers Initiative, Alvim, M.K.M., Ames, D., Anderson, T.J., Andreassen, O.A., Arias-Vasquez, A., Bastin, M.E., Baune, B.T., Beckham, J.C., Blangero, J., Boomsma, D.I., Brodaty, H., Brunner, H.G., Buckner, R.L., Buitelaar, J. K., Bustillo, J.R., Cahn, W., Cairns, M.J., Calhoun, V., Carr, V.J., Caseras, X., Caspers, J., Cavalleri, G.L., Cendes, F., Corvin, A., Crespo-Facorro, B., Dalrymple-Alford, J.C., Dannlowski, U., de Geus, E.J.C., Deary, I.J., Delanty, N., Depondt, C., Desrivieres, S., Donohoe, G., Espeseth, T., Fernández, G., Fisher, S.E., Flor, H., Forstner, A.J., Francks, C., Franke, B., Glahn, D.C., Gollub, R.L., Grabe, H.J., Gruber, O., Häberg, A.K., Hariri, A.R., Hartman, C.A., Hashimoto, R., Heinz, A., Henskens, F.A., Hillegers, M.H.J., Hoekstra, P.J., Holmes, A.J., Hong, L.E., Hopkins, W.D., Hulshoff Pol, H.E., Jernigan, T.L., Jönsson, E.G., Kahn, R.S., Kennedy, M.A., Kircher, T.T.J., Kochunov, P., Kwok, J.B.J., Le Hellard, S., Loughland, C.M., Martin, N.G., Martinot, J.-L., McDonald, C., McMahon, K.L., Meyer-Lindenberg, A., Michie, P.T., Morey, R.A., Mowry, B., Nyberg, L., Oosterlaan, J., Ophoff, R.A., Patalis, C., Paus, T., Pausova, Z., Penninx, B.W.J.H., Polderman, T.J.C., Posthuma, D., Rietschel, M., Roffman, J.L., Rowland, L.M., Sachdev, P.S., Sämann, P.G., Schall, U., Schumann, G., Scott, R.J., Sim, K., Sisodiya, S.M., Smoller, J.W., Sommer, I.E., St Pourcain, B., Stein, D.J., Toga, A.W., Trollor, J.N., Van der Wee, N.J.A., van ’t Ent, D., Völzke, H., Walter, H., Weber, B., Weinberger, D.R., Wright, M.J., Zhou, J., Stein, J.L., Thompson, P.M., Medland, S.E., Enhancing Neuromaging Genetics through Meta-Analysis Consortium (ENIGMA)—Genetics working group, 2020. The genetic architecture of the human cerebral cortex. *Science* 367, eaay6690. <https://doi.org/10.1126/science.aay6690>.
- Hagler, D.J., Hattori, S., Cornejo, M.D., Makowski, C., Fair, D.A., Dick, A.S., Sutherland, M.T., Casey, B.J., Barch, D.M., Harms, M.P., Watts, R., Bjork, J.M., Garavan, H.P., Hilmer, L., Pung, C.J., Scat, C.S., Kuperman, J., Bartsch, H., Xue, F., Heitzeg, M.M., Laird, A.R., Trinh, T.T., Gonzalez, R., Tapert, S.F., Riedel, M.C., Squeglia, L.M., Hyde, L.W., Rosenberg, M.D., Earl, E.A., Howlett, K.D., Baker, F.C., Soules, M., Diaz, J., de Leon, O.R., Thompson, W.K., Neale, M.C., Herting, M., Sowell, E.R., Alvarez, R.P., Hawes, S.W., Sanchez, M., Bodurka, J., Breslin, F.J., Morris, A.S., Paulus, M.P., Simmons, W.K., Polimeni, J.R., van der Kouwe, A., Nencka, A.S., Gray, K.M., Pierpaoli, C., Matochik, J.A., Noronha, A., Aklin, W.M., Conway, K., Glantz, M., Hoffman, E., Little, R., Lopez, M., Pariyadath, V., Weiss, S.R., Wolff-Hughes, D.L., DelCarmen-Wiggins, R., Feldstein Ewing, S.W., Miranda-Dominguez, O., Nagel, B.J., Perrone, A.J., Sturgeon, D.T., Goldstone, A., Pfefferbaum, A., Pohl, K.M., Prouty, D., Uban, K., Bookheimer, S.Y., Dapretto, M., Galvan, A., Bagot, K., Giedd, J., Infante, M.A., Jacobus, J., Patrick, K., Shilling, P.D., Desikan, R., Li, Y., Sugrue, L., Banich, M.T., Friedman, N., Hewitt, J.K., Hopfer, C., Sakai, J., Tanabe, J., Cottler, L.B., Nixon, S.J., Chang, L., Cloak, C., Ernst, T., Reeves, G., Kennedy, D.N., Heeringa, S., Peltier, S., Schulenberg, J., Siprda, C., Zucker, R.A., Iacono, W.G., Luciano, M., Calabro, F.J., Clark, D.B., Lewis, D.A., Luna, B., Schirda, C., Brima, T., Foxe, J.J., Freedman, E.G., Mruzek, D.W., Mason, M. J., Huber, R., McGlade, E., Prescott, A., Renshaw, P.F., Yurgelun-Todd, D.A., Allgaier, N.A., Dumas, J.A., Ivanova, M., Potter, A., Florsheim, P., Larson, C., Lisdahl, K., Charness, M.E., Fuemmeler, B., Hetttema, J.M., Maes, H.H., Steinberg, J., Anokhin, A.P., Glaser, P., Heath, A.C., Madden, P.A., Baskin-Sommers, A., Constable, R.T., Grant, S.J., Dowling, G.J., Brown, S.A., Jernigan, T.L., Dale, A.M., 2019. Image processing and analysis methods for the Adolescent Brain Cognitive Development Study. *Neuroimage* 202, 116091. <https://doi.org/10.1016/j.neuroimage.2019.116091>.
- Hare, T.A., Camerer, C.F., Rangel, A., 2009. Self-control in decision-making involves modulation of the vmPFC valuation system. *Science* 324, 646–648. <https://doi.org/10.1126/science.1168450>.
- Hoogman, M., Muetzel, R., Guimaraes, J.P., Shumskaya, E., Mennes, M., Zwiers, M.P., Jahanshad, N., Sudre, G., Wolfers, T., Earl, E.A., Soliva Vila, J.C., Vives-Gilabert, Y., Khadka, S., Novotny, S.E., Hartman, C.A., Heslenfeld, D.J., Schwenen, L.J.S., Ambrosino, S., Oranje, B., de Zeeuw, P., Chaim-Avancini, T.M., Rosa, P.G.P., Zanetti, M.V., Malpas, C.B., Kohls, G., von Polier, G.G., Seitz, J., Biederman, J., Doyle, A.E., Dale, A.M., van Erp, T.G.M., Epstein, J.N., Jernigan, T.L., Baur-Streubel, R., Ziegler, G.C., Zierhut, K.C., Schranter, A., Høvik, M.F., Lundervold, A.J., Kelly, C., McCarthy, H., Skokauskas, N., O’Gorman Tuura, R.L., Calvo, A., Lera-Miguel, S., Nicolau, R., Chantiluke, K.C., Christakou, A., Vance, A., Cercignani, M., Gabel, M.C., Asherson, P., Baumeister, S., Brandeis, D., Hohmann, S., Bramati, I.E., Tovar-Moll, F., Fallgatter, A.J., Kardatzki, B., Schwarz, L., Anikin, A., Baranov, A.,

- Gogberashvili, T., Kapilushniy, D., Solovieva, A., El Marroun, H., White, T., Karkashadze, G., Namazova-Baranova, L., Ethofer, T., Mattos, P., Banaschewski, T., Coghill, D., Plessen, K.J., Kuntsi, J., Mehta, M.A., Paloyelis, Y., Harrison, N.A., Bellgrove, M.A., Silk, T.J., Cubillo, A.I., Rubia, K., Lázaro, L., Brem, S., Walitza, S., Frodl, T., Zentis, M., Castellanos, F.X., Yoncheva, Y.N., Haavik, J., Reneman, L., Conzelmann, A., Lesch, K.-P., Pauli, P., Reif, A., Tamm, L., Konrad, K., Oberwilleand Weiss, E., Busatto, G.F., Louza, M.R., Durston, S., Hoeksma, P.J., Oosterlaan, J., Stevens, M.C., Ramos-Quiroga, J.A., Vilarroya, O., Fair, D.A., Nigg, J.T., Thompson, P.M., Buitelaar, J.K., Faraone, S.V., Shaw, P., Tiemeier, H., Bralten, J., Franke, B., 2019. Brain imaging of the cortex in ADHD: a coordinated analysis of large-scale clinical and population-based samples. *AJP* 176, 531–542. <https://doi.org/10.1176/appi.ajp.2019.18091033>.
- Japee, S., Holaday, K., Satyshur, M.D., Mukai, I., Ungerleider, L.G., 2015. A role of right middle frontal gyrus in reorienting of attention: a case study. *Front. Syst. Neurosci.* 9. <https://doi.org/10.3389/fnsys.2015.00023>.
- Jernigan, T.L., Brown, S.A., Dale, A.M., Tapert, S.F., Sowell, E.R., Herting, M., Laird, A., Gonzalez, R., Squeglia, L., Gray, K., Paulus, M.P., Aupperle, R., Feldstein Ewing, S. W., Nagel, B.J., Fair, D.A., Baker, F., Müller Oehring, E., Bookheimer, S.Y., Dapretto, M., Jacobus, J., Wilson, S., Banich, M.T., Cottler, L.B., Nixon, S.J., Ernst, T.M., Chang, L., Heitzeg, M.M., Sripada, C., Luciana, M.M., Friedman, N., Clark, D.B., Luna, B., Foxe, J., Freedman, E., Yurgelun-Todd, D.A., Renshaw, P.F., Potter, A., Garavan, H.P., Lisdahl, K., Larson, C., Bjork, J.M., Neale, M.C., Heath, A.C., Barch, D. M., Madden, P.A., Casey, B.J., Baskin-Sommers, A., Gee, D., 2025. ABCD study(R) data release 6.0. <https://doi.org/10.82525/JY7N-G441>.
- Jirsaraie, R.J., Gatavins, M.M., Pines, A.R., Kandala, S., Bijsterbosch, J.D., Marek, S., Bogdan, R., Barch, D.M., Sotiras, A., 2025. Mapping the neurodevelopmental predictors of psychopathology. *Mol. Psychiat.* 30, 478–488. <https://doi.org/10.1038/s41380-024-02682-7>.
- Khalaf, Abderrahman M., Alubied, A.A., Khalaf, Ahmed M., Rifaey, A.A., 2023. The impact of social media on the mental health of adolescents and young adults: a systematic review. *Cureus* 15, e42990. <https://doi.org/10.7759/cureus.42990>.
- Kollndorfer, K., Novak, A., Nenning, K.-H., Fischmeister, F.P.S., Seidl, R., Langs, G., Kasprian, G., Prayer, D., Bartha-Doering, L., 2023. Cortical thickness in the right medial frontal gyrus predicts planning performance in healthy children and adolescents. *Front. Psychol.* 14, 1196707. <https://doi.org/10.3389/fpsyg.2023.1196707>.
- Kucyi, A., Esterman, M., Riley, C.S., Valera, E.M., 2016. Spontaneous default network activity reflects behavioral variability independent of mind-wandering. *Proc. Natl. Acad. Sci.* 113, 13899–13904. <https://doi.org/10.1073/pnas.1611743113>.
- Kuehne, M., Schmidt, K., Heinze, H.-J., Zaehele, T., 2019. Modulation of emotional conflict processing by high-definition transcranial direct current stimulation (HD-TDCS). *Front. Behav. Neurosci.* 13. <https://doi.org/10.3389/fnbeh.2019.00224>.
- Lee, Y., Chahal, R., Gotlib, I.H., 2024. The default mode network is associated with changes in internalizing and externalizing problems differently in adolescent boys and girls. *Dev. Psychopathol.* 36, 834–843. <https://doi.org/10.1017/S0954579423000111>.
- Marsh, R., Gerber, A.J., Peterson, B.S., 2008. Neuroimaging studies of normal brain development and their relevance for understanding childhood neuropsychiatric disorders. *J. Am. Acad. Child Adolesc. Psychiat.* 47, 1233–1251. <https://doi.org/10.1097/CHI.0b013e318185e703>.
- McAlister, K.L., Beatty, C.C., Smith-Caswell, J.E., Yourell, J.L., Huberty, J.L., 2024. Social media use in adolescents: bans, benefits, and emotion regulation behaviors. *JMIR Ment. Health* 11, e64626. <https://doi.org/10.2196/64626>.
- Michelin, G., Barch, D.M., Tian, Y., Watson, D., Klein, D.N., Kotov, R., 2019. Delineating and validating higher-order dimensions of psychopathology in the Adolescent Brain Cognitive Development (ABCD) Study. *Transl. Psychiat.* 9, 261. <https://doi.org/10.1038/s41380-019-0593-4>.
- Miller, E.K., Cohen, J.D., 2001. An integrative theory of prefrontal cortex function. *Annu. Rev. Neurosci.* 24, 167–202. <https://doi.org/10.1146/annurev.neuro.24.1.167>.
- Miller, J., Mills, K.L., Vuore, M., Orben, A., Przybylski, A.K., 2023. Impact of digital screen media activity on functional brain organization in late childhood: evidence from the ABCD Study. *Cortex* 169, 290–308. <https://doi.org/10.1016/j.cortex.2023.09.009>.
- Muppalla, S.K., Vuppapalati, S., Reddy Pulliahgaru, A., Sreenivasulu, H., 2023. Effects of excessive screen time on child development: an updated review and strategies for management. *Cureus* 15, e40608. <https://doi.org/10.7759/cureus.40608>.
- Nagata, J.M., Cortez, C.A., Iyer, P., Ganson, K.T., Chu, J., Conroy, A.A., 2022. Parent-adolescent discrepancies in adolescent recreational screen time reporting during the Coronavirus disease 2019 pandemic. *Acad. Pediatr.* 22, 413–421. <https://doi.org/10.1016/j.acap.2021.12.008>.
- Nagata, J.M., Memon, Z., Talebloo, J., Li, K., Low, P., Shao, I.Y., Ganson, K.T., Testa, A., He, J., Brindis, C.D., Baker, F.C., 2025. Prevalence and patterns of social media use in early adolescents. *Acad. Pediatr.* 25, 102784. <https://doi.org/10.1016/j.acap.2025.102784>.
- Nivins, S., Sauce, B., Liebherr, M., Judd, N., Klingberg, T., 2024. Long-term impact of digital media on brain development in children. *Sci. Rep.* 14, 13030. <https://doi.org/10.1038/s41598-024-63566-y>.
- Oehrn, C.R., Hanslmayr, S., Fell, J., Deuker, L., Kremers, N.A., Lam, A.T.D., Elger, C.E., Axmacher, N., 2014. Neural communication patterns underlying conflict detection, resolution, and adaptation. *J. Neurosci.* 34, 10438–10452. <https://doi.org/10.1523/JNEUROSCI.3099-13.2014>.
- Office of the Surgeon General, 2024. Social media and youth mental health [WWW Document]. URL: <https://www.hhs.gov/surgeongeneral/reports-and-publications/youth-mental-health/social/index.html> (accessed 2.19.25).
- Padmanabhan, A., Lynch, C.J., Schaer, M., Menon, V., 2017. The default mode network in autism. *Biol. Psychiat. Cogn. Neurosci. Neuroimaging* 2, 476–486. <https://doi.org/10.1016/j.bpsc.2017.04.004>.
- Palmer, C.E., Pecheva, D., Iversen, J.R., Hagler, D.J., Sugrue, L., Nedelec, P., Fan, C.C., Thompson, W.K., Jernigan, T.L., Dale, A.M., 2022. Microstructural development from 9 to 14 years: evidence from the ABCD Study. *Dev. Cogn. Neurosci.* 53, 101044. <https://doi.org/10.1016/j.dcn.2021.101044>.
- Panizzon, M.S., Fennema-Notestine, C., Eyer, L.T., Jernigan, T.L., Prom-Wormley, E., Neale, M., Jacobson, K., Lyons, M.J., Grant, M.D., Franz, C.E., Xian, H., Tsuang, M., Fischl, B., Seidman, L., Dale, A., Kremen, W.S., 2009. Distinct genetic influences on cortical surface area and cortical thickness. *Cereb. Cortex* 19, 2728–2735. <https://doi.org/10.1093/cercor/bhp026>.
- Parekh, P., Fan, C.C., Frei, O., Palmer, C.E., Smith, D.M., Makowski, C., Iversen, J.R., Pecheva, D., Holland, D., Loughnan, R., Nedelec, P., Thompson, W.K., Hagler, D.J., Andreassen, O.A., Jernigan, T.L., Nichols, T.E., Dale, A.M., 2024. FEMA: Fast and Efficient Mixed-effects Algorithm for large sample whole-brain imaging data. *Hum. Brain Mapp.* 45, e26579. <https://doi.org/10.1002/hbm.26579>.
- Paulus, M.P., Squeglia, L.M., Bagot, K., Jacobus, J., Kuplicki, R., Breslin, F.J., Bodurka, J., Morris, A.S., Thompson, W.K., Bartsch, H., Tapert, S.F., 2019. Screen media activity and brain structure in youth: evidence for diverse structural correlation networks from the ABCD Study. *NeuroImage* 185, 140–153. <https://doi.org/10.1016/j.neuroimage.2018.10.040>.
- Pehlivanova, M., Wolf, D.H., Sotiras, A., Kaczurkin, A.N., Moore, T.M., Ciric, R., Cook, P.A., Garza, A.G., de, L., Rosen, A.F.G., Ruparel, K., Sharma, A., Shinohara, R. T., Roalf, D.R., Gur, R.C., Davatzikos, C., Gur, R.E., Kable, J.W., Satterthwaite, T.D., 2018. Diminished cortical thickness is associated with impulsive choice in adolescence. *J. Neurosci.* 38, 2471–2481. <https://doi.org/10.1523/JNEUROSCI.2200-17.2018>.
- Raichle, M.E., 2015. The brain's default mode network. *Annu. Rev. Neurosci.* 38, 433–447. <https://doi.org/10.1146/annurev-neuro-071013-014030>.
- Rauschecker, A.M., Nedelec, P., Pan, S., Olaru, M., Nillo, R.M., Palmer, C.E., Pecheva, D., Dale, A.M., Jernigan, T.L., Sugrue, L.P., 2025. Neurocognitive and brain structure correlates of reading and television habits in early adolescence. *Sci. Rep.* 15, 6235. <https://doi.org/10.1038/s41598-025-88398-2>.
- Schmaal, L., Hibar, D.P., Sämann, P.G., Hall, G.B., Baune, B.T., Jahanshad, N., Cheung, J. W., van Erp, T.G.M., Bos, D., Ikram, M.A., Vernooij, M.W., Niessen, W.J., Tiemeier, H., Hofman, A., Wittfeld, K., Grabe, H.J., Janowitz, D., Bülow, R., Selonke, M., Völzke, H., Grotegerd, D., Dannlowski, U., Arolt, V., Opel, N., Heindel, W., Kugel, H., Hoehn, D., Czisch, M., Couvy-Duchesne, B., Rentería, M.E., Strike, L.T., Wright, M.J., Mills, N.T., de Zubicaray, G.L., McMahon, K.L., Medland, S. E., Martin, N.G., Gillespie, N.A., Goya-Maldonado, R., Gruber, O., Krämer, B., Hatton, S.N., Lagopoulos, J., Hickie, I.B., Frodl, T., Carballo, A., Frey, E.M., van Velzen, L.S., Penninx, B.W.J.H., van Tol, M.-J., van der Wee, N.J., Davey, C.G., Harrison, B.J., Mwambi, B., Cao, B., Soares, J.C., Veer, I.M., Walter, H., Schoepf, D., Zuroski, B., Konrad, C., Schramm, E., Normann, C., Schnell, K., Sacchet, M.D., Gotlib, I.H., MacQueen, G.M., Godlewski, B.R., Nickson, T., McIntosh, A.M., Pampmeyer, M., Whalley, H.C., Hall, J., Sussmann, J.E., Li, M., Walter, M., Aftanas, L., Brack, I., Bokhan, N.A., Thompson, P.M., Veltman, D.J., 2017. Cortical abnormalities in adults and adolescents with major depression based on brain scans from 20 cohorts worldwide in the ENIGMA major depressive disorder working group. *Mol. Psychiat.* 22, 900–909. <https://doi.org/10.1038/mp.2016.60>.
- Shaw, P., Greenstein, D., Lerch, J., Clasen, L., Lenroot, R., Gogtay, N., Evans, A., Rapoport, J., Giedd, J., 2006. Intellectual ability and cortical development in children and adolescents. *Nature* 440, 676–679. <https://doi.org/10.1038/nature04513>.
- Sotiras, A., Toledo, J.B., Gur, R.E., Gur, R.C., Satterthwaite, T.D., Davatzikos, C., 2017. Patterns of coordinated cortical remodeling during adolescence and their associations with functional specialization and evolutionary expansion. *Proc. Natl. Acad. Sci.* 114, 3527–3532. <https://doi.org/10.1073/pnas.1620928114>.
- Squeglia, L.M., Jacobus, J., Sorg, S.F., Jernigan, T.L., Tapert, S.F., 2013. Early adolescent cortical thinning is related to better neuropsychological performance. *J. Int. Neuropsychol. Soc.* 19, 962–970. <https://doi.org/10.1017/S1355617713000878>.
- Stevens, F.L., Hurler, R.A., Taber, K.H., 2011. Anterior cingulate cortex: unique role in cognition and emotion. *J. Neuropsychiat. Clin. Neurosci.* 23, 121–125. <https://doi.org/10.1176/jnp.23.2.jnp121>.
- Tamnes, C.K., Herting, M.M., Goddings, A.-L., Meuwese, R., Blakemore, S.-J., Dahl, R.E., Güroğlu, B., Raznahan, A., Sowell, E.R., Crone, E.A., Mills, K.L., 2017. Development of the cerebral cortex across adolescence: a multisample study of inter-related longitudinal changes in cortical volume, surface area, and thickness. *J. Neurosci.* 37, 3402–3412. <https://doi.org/10.1523/JNEUROSCI.3302-16.2017>.
- van Druenen, L., Dobbelaar, S., Crone, E.A., Wierenga, L.M., 2024. Genetic and environmental influences on structural brain development from childhood to adolescence: a longitudinal twin study on cortical thickness, surface area, and subcortical volume. *Dev. Cogn. Neurosci.* 68, 101407. <https://doi.org/10.1016/j.dcn.2024.101407>.
- Volkow, N.D., Koob, G.F., Croyle, R.T., Bianchi, D.W., Gordon, J.A., Koroshetz, W.J., Pérez-Stable, E.J., Riley, W.T., Bloch, M.H., Conway, K., Deeds, B.G., Dowling, G.J., Grant, S., Howlett, K.D., Matochik, J.A., Morgan, G.D., Murray, M.M., Noronha, A., Spong, C.Y., Wargo, E.M., Warren, K.R., Weiss, S.R.B., 2018. The conception of the ABCD Study: from substance use to a broad NIH collaboration. *Dev. Cogn. Neurosci.* 32, 4–7. <https://doi.org/10.1016/j.dcn.2017.10.002>.
- Wang, Y., Ma, L., Wang, J., Ding, Y., Men, W., Tan, S., Gao, J.-H., Qin, S., He, Y., Dong, Q., Tao, S., 2025. Connections between the middle frontal gyrus and the dorsolateral attention network are associated with the development of attentional symptoms. *Biol. Psychiat.* 97, 531–539. <https://doi.org/10.1016/j.biopsych.2024.04.019>.

- Wierenga, L.M., Langen, M., Oranje, B., Durston, S., 2014. Unique developmental trajectories of cortical thickness and surface area. *NeuroImage* 87, 120–126. <https://doi.org/10.1016/j.neuroimage.2013.11.010>.
- Williams, V.J., Juranek, J., Cirino, P., Fletcher, J.M., 2018. Cortical thickness and local gyrification in children with developmental dyslexia. *Cereb Cortex* 28, 963–973. <https://doi.org/10.1093/cercor/bhx001>.
- Winkler, A.M., Kochunov, P., Blangero, J., Almasy, L., Zilles, K., Fox, P.T., Duggirala, R., Glahn, D.C., 2010. Cortical thickness or grey matter volume? The importance of selecting the phenotype for imaging genetics studies. *NeuroImage Imag. Genet.* 53, 1135–1146. <https://doi.org/10.1016/j.neuroimage.2009.12.028>.
- Xiao, Q., Fu, Y., Yi, X., Ding, J., Han, Z., Zhang, Z., Tan, Z., Wang, J., Wu, Z., Pi, J., Chen, B.T., 2023. Altered cortical thickness and emotional dysregulation in adolescents with borderline personality disorder. *Eur. J. Psychotraumatol.* 14, 2163768. <https://doi.org/10.1080/20008066.2022.2163768>.
- Yadav, A., Purushotham, A., 2025. Cortical structure in nodes of the default mode network estimates general intelligence. *Brain Behav.* 15, e70531. <https://doi.org/10.1002/brb3.70531>.
- Yamagishi, T., Takagishi, H., Fermin, A., de, S.R., Kanai, R., Li, Y., Matsumoto, Y., 2016. Cortical thickness of the dorsolateral prefrontal cortex predicts strategic choices in economic games. *Proc. Natl. Acad. Sci. U S A* 113, 5582–5587. <https://doi.org/10.1073/pnas.1523940113>.
- Yan, C.-G., Chen, X., Li, L., Castellanos, F.X., Bai, T.-J., Bo, Q.-J., Cao, J., Chen, G.-M., Chen, N.-X., Chen, W., Cheng, C., Cheng, Y.-Q., Cui, X.-L., Duan, J., Fang, Y.-R., Gong, Q.-Y., Guo, W.-B., Hou, Z.-H., Hu, L., Kuang, L., Li, F., Li, K.-M., Li, T., Liu, Y.-S., Liu, Z.-N., Long, Y.-C., Luo, Q.-H., Meng, H.-Q., Peng, D.-H., Qiu, H.-T., Qiu, J., Shen, Y.-D., Shi, Y.-S., Wang, C.-Y., Wang, F., Wang, K., Wang, L., Wang, X., Wang, Y., Wu, X.-P., Wu, X.-R., Xie, C.-M., Xie, G.-R., Xie, H.-Y., Xie, P., Xu, X.-F., Yang, H., Yang, J., Yao, J.-S., Yao, S.-Q., Yin, Y.-Y., Yuan, Y.-G., Zhang, A.-X., Zhang, H., Zhang, K.-R., Zhang, L., Zhang, Z.-J., Zhou, R.-B., Zhou, Y.-T., Zhu, J.-J., Zou, C.-J., Si, T.-M., Zuo, X.-N., Zhao, J.-P., Zang, Y.-F., 2019. Reduced default mode network functional connectivity in patients with recurrent major depressive disorder. *Proc. Natl. Acad. Sci.* 116, 9078–9083. <https://doi.org/10.1073/pnas.1900390116>.
- Zhao, Y., Paulus, M., Bagot, K.S., Constable, R.T., Yaggi, H.K., Redeker, N.S., Potenza, M. N., 2022. Brain structural covariation linked to screen media activity and externalizing behaviors in children. *J. Behav. Addict.* 11, 417–426. <https://doi.org/10.1556/2006.2022.00044>.
- Zhao, Y., Paulus, M.P., Potenza, M.N., 2023. Brain structural co-development is associated with internalizing symptoms two years later in the ABCD cohort. *J. Behav. Addict.* 12, 80–93. <https://doi.org/10.1556/2006.2023.00006>.

Glossary

ABCD: Adolescent Brain Cognitive Development
ADHD: Attention-deficit/hyperactivity disorder
DMN: Default mode network
FDR: False discovery rate
FEMA: Fast and Efficient Mixed Effects Algorithm
fMRI: Functional magnetic resonance imaging
MRI: Magnetic resonance imaging
rMFG: Right middle frontal gyrus
ROI: Region-of-Interest
SD: Standard deviation
sMRI: Structural magnetic resonance imaging